TokTickIT: Actions Taken, Dashboards, and Concurrency Control

Lab 4 Engineering Report & Verification Evidence Package (CPE 334)

PULL REQUEST (PR #55)
https://github.com/pimchayasupr-hash/toktickit/pull/55

BRANCH & INTEGRATION
feature/lab4-hardening → lab4-staging → main

AUTHOR & REVIEWER
Author: @pimchayasupr-hash • Reviewer: @MiMikoChAn913

AUTOMATED VERIFICATION STATUS
100% PASSED (144 SERVER + 40 CLIENT + 11 E2E)

1. Executive Summary & Reviewer-Driven Hardening

TokTickIT Lab 4 extends the enterprise IT Service Desk with full operational task tracking (**Actions Taken**), strict transition gates (**Resolution Gate**), database-level concurrency protection (**Atomic Optimistic Locking with Predicates**), immutable audit trails (**Ticket Status History**), and role-tailored real-time **Dashboards**.

Following rigorous code review by `@MiMikoChAn913` on Pull Request #55, 5 critical areas (4 × P1, 1 × P2) were hardened with 8 new regression tests:

- **Action Lifecycle & Assignee**: Implemented `ActionStatus` enum (`PENDING` , `IN_PROGRESS` , `COMPLETED` , `CANCELLED`), operational assignee delegation (`assigneeId`), inactive-assignee rejection (`INACTIVE_ASSIGNEE`), terminal action immutability, and conditional result enforcement.
- **Atomic Optimistic Concurrency**: Replaced in-memory checks with atomic database write predicates `updateMany({ where: { id, version } })` inside transactions, returning HTTP 409 Conflict without silent overwrites.
- **Mandatory Version Validation**: Enforced numeric version validation on status writes (HTTP 400 on missing version) and atomic compare-and-increment on tickets.
- **Append-Only Ticket Status History**: Added `TicketStatusHistory` model, writing exactly 1 actor-attributed row in the same transaction as successful status updates, and 0 rows on conflicts.
- **Deterministic E2E Resolution Gate Verification**: Eliminated conditional skips, asserting exact rejection on fixture `TKT-2026-001233` and resolution success with audit trail on `TKT-2026-001234` .

2. Peer Review Verification Matrix (PR #55)

Priority	Reviewer Comment (@MiMikoChAn913)	Technical Resolution & Implementation	Regression Test Evidence
[P1] LIFECYCLE	Missing ActionTaken lifecycle, operational assigneeId, inactive-assignee rejection, and premature result requirement.	Added ActionStatus enum, assigneeId relation to User, validateAssignee check, terminal immutability, and made result required only on COMPLETED.	`TEST-ACT-18..20`, `TEST-ACT-22`, `TEST-UI-ACT-04..05`, `E2E-ACT-02` PASS
[P1] ACTION LOCK	Non-atomic optimistic concurrency permits silent lost updates under concurrent writes.	Implemented atomic database predicate `updateMany({ where: { id, version } })` in transaction; returns 409 Conflict if count is 0.	`TEST-ACT-21` (Concurrent write race condition) PASS
[P1] TICKET LOCK	Omitting version bypasses ticket concurrency, and update by id permits race conditions.	Enforced numeric version validation (400 Bad Request) and atomic compare-and-increment `updateMany` in database transaction.	`TEST-WFK-06`, `TEST-WFK-07` (Stale write & race condition) PASS
[P1] HISTORY	Status change updates mutable ticket only without append-only TicketStatusHistory.	Created TicketStatusHistory table; appended actor-attributed row atomically in same transaction (0 rows appended on conflict).	`TEST-WFK-08`, `GET /api/staff/tickets/:id/status-history` PASS
[P2] E2E GATE	Conditional `if (canSelectResolved)` allows E2E test to pass without exercising resolution gate.	Used deterministic fixtures `TKT-2026-001233` (0 actions → rejected) and `TKT-2026-001234` (completed → resolved with history).	`E2E-RES-01`, `E2E-RES-02` (Deterministic) PASS

3. Visual Evidence: Dashboards & Responsive Layouts

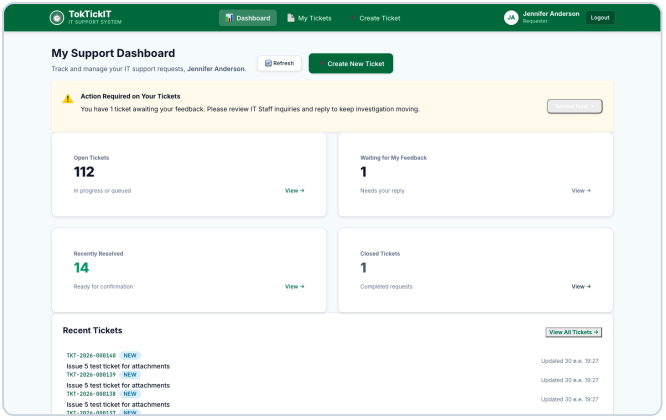


Fig 1a: Requester Dashboard (Desktop 1440×900) — Status Cards & Attention Banner

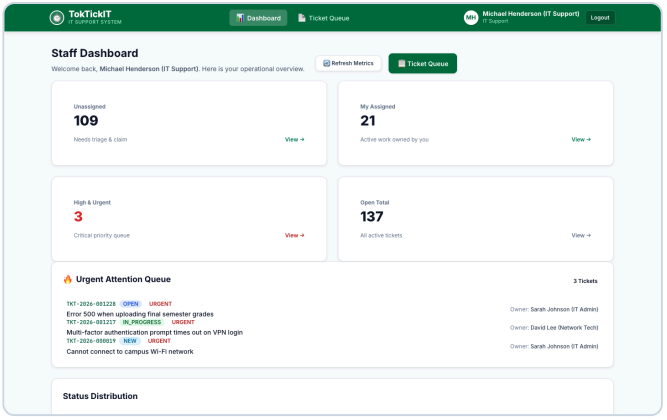


Fig 1b: IT Staff Dashboard (Desktop 1440×900) — Team Queue, Status & Priority Breakdown

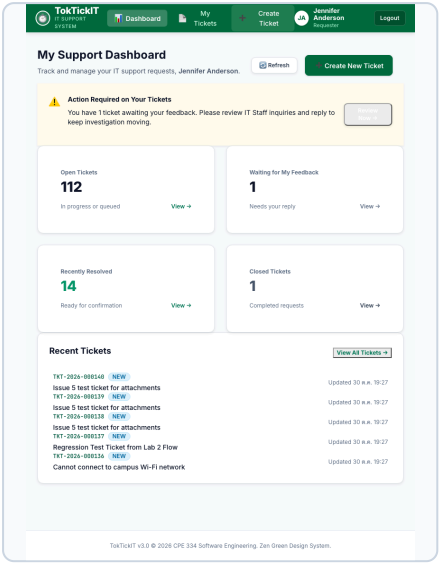


Fig 2a: Requester Dashboard (Tablet 820×1180)

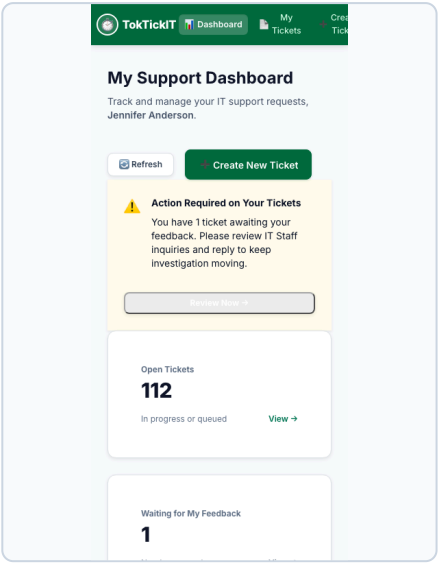


Fig 2b: Requester Dashboard (Mobile 375×812)

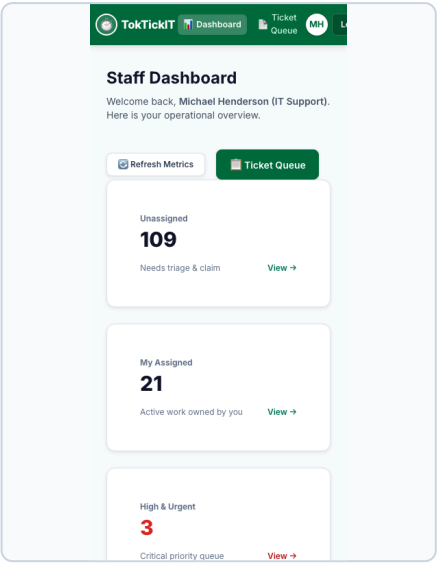


Fig 2c: Staff Dashboard (Mobile 375×812)

4. Visual Evidence: Actions Taken Lifecycle, Resolution Gate & Audit History

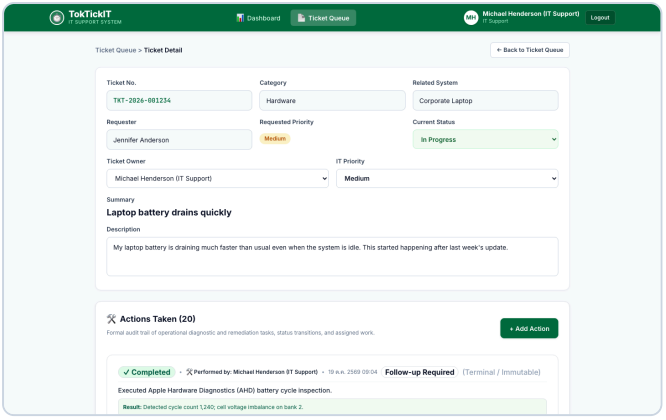


Fig 3a: Actions Taken Subsystem — Badges (Pending, In Progress, Completed), Assignee, and Terminal Lock

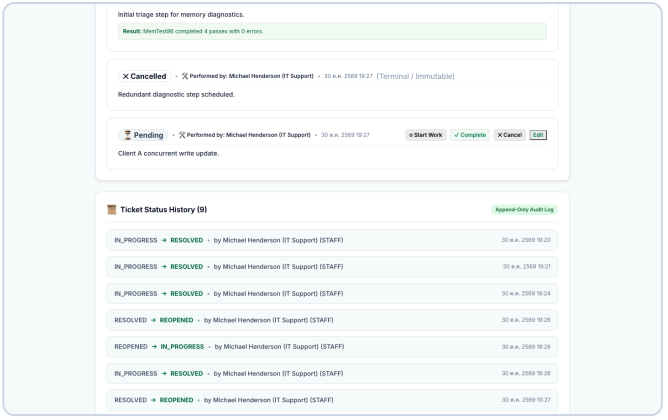


Fig 3b: Append-Only Ticket Status History — Actor-Attributed Audit Trail

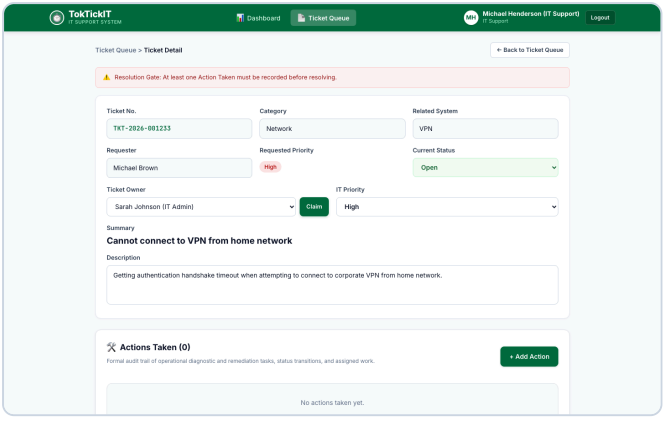


Fig 4a: Resolution Gate Rejection Alert — Blocks transition when 0 actions recorded

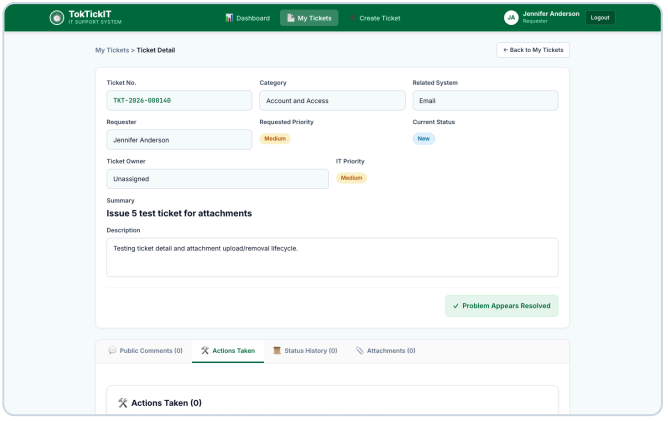


Fig 4b: Requester Read-Only View — Visibility into Actions Taken and Status History without Edit controls

5. Comprehensive Test Execution Report

Traceability Matrix Verification: 100% of defined Acceptance Criteria (AC-01..AC-23), Functional Requirements (FR-01..FR-21), and Business Rules (BR-01..BR-18) are covered without orphan references.

Suite	Framework	Tests Run	Passed	Failed	Duration
Traceability Script	Python 3 (AST Parser)	62 Checks	62	0	0.12s
Server API & Concurrency Suites	Vitest + Supertest	144 Tests (29 files)	144	0	24.42s
Client Component & Dashboard Suites	Vitest + RTL	40 Tests (17 files)	40	0	8.67s
Playwright Multi-Browser E2E Suite	Playwright + Axe-core	11 Tests (3 files)	11	0	17.70s
TOTAL AUTOMATED AUDIT		**257 Checks**	**257 (100%)**	**0 (0%)**	**50.91s**

Verification Commands:

```
npm run test:traceability
npm run test:server
npm run test:client
npx playwright test e2e/lab-04/ --project=chromium
```